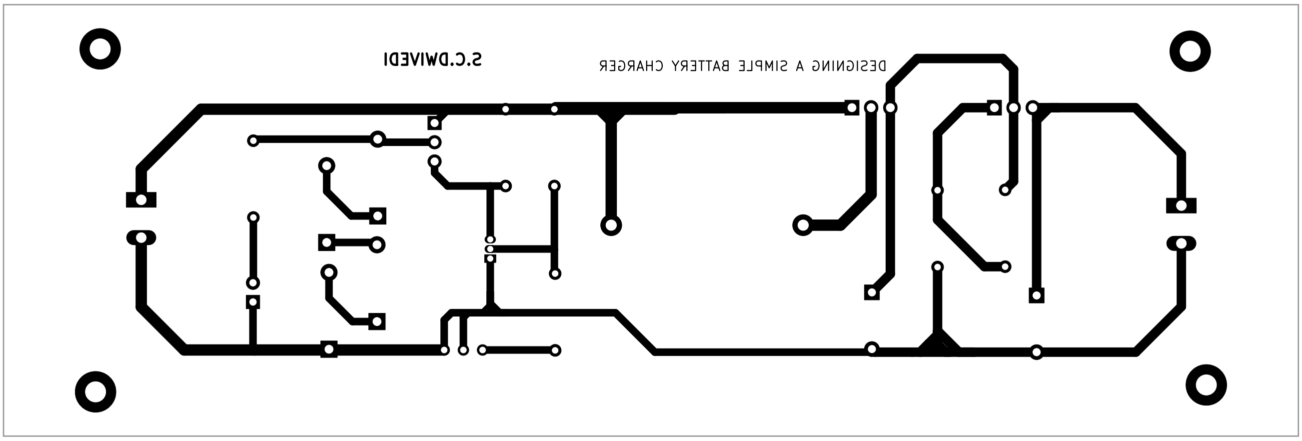

Fig. 3: Actual-size PCB layout for the circuit

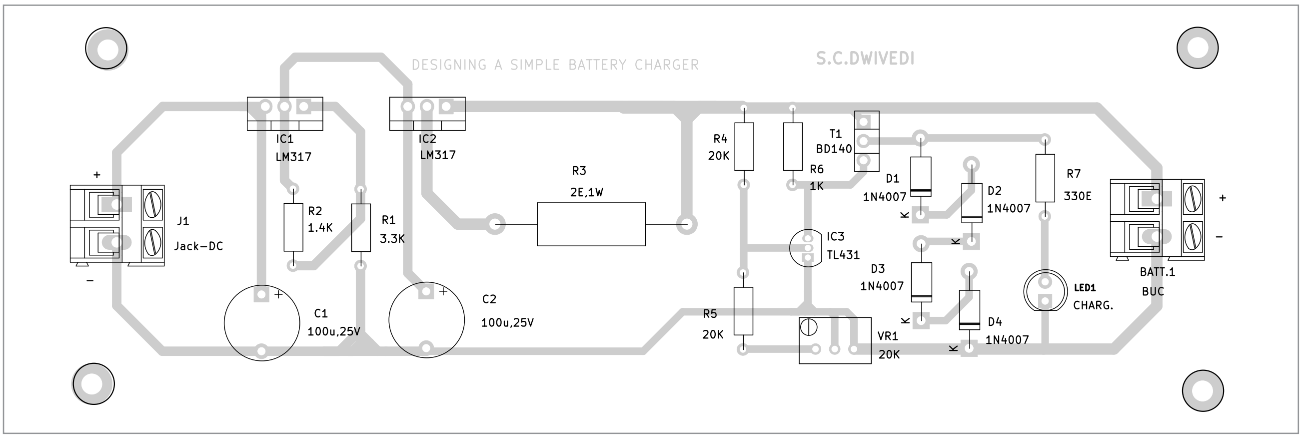

Fig. 4: Component layout of the PCB

series with four diodes, which act as a load. The diodes are also connected to a resistor and an LED in parallel to them. When the transistor is conducting, the current flows through the four diodes and the LED simultaneously, thus turning on the LED to indicate that the battery is fully charged.

The circuit also offers a cell balancing feature, which is important when we charge a battery with multiple cells in series. With the cells connected in series, we need to make sure that the total voltage of the battery pack after charging does not exceed the maximum specified voltage of the battery pack. Also, the voltage of each charged cell should not exceed the maximum specified voltage of that individual cell.

Current limiter circuit

Every cell has a maximum limit of charging current it can take, which is denoted by its C-rate. The C-rate of a cell depends on multiple factors, such as its chemistry, size, internal structure, etc. Charging with overcurrent may cause irreplaceable damage to the cell and may also cause thermal runaway in the battery, which can result in a fire. Therefore, IC LM317 is used as a current limiter.

As shown in Fig. 2, the voltage input pin VI of LM317 is connected to the positive of the source, and voltage output pin VO is connected to resistor R3. The adjust pin ADJ is connected to the other end of the resistor. The value of current I_out can be adjusted by changing the value of resistor R3, as per the relationship given below. Though, for this charger, we will keep the maximum output current (Iout) as 0.6A.

$$Iout = \frac{V_{ref}}{R3}$$

Here, V_{ref} is 1.25V.

$$0.6 = \frac{1.25}{R3}$$

R3 is 2.08 ohm

Constant voltage source

To make the circuit more versatile and make it work over a wide voltage range, we employ voltage control by using another LM317 IC. The input terminal VI of this IC is connected to the ground through capacitor C1, which should be kept as close as possible to the input ter-